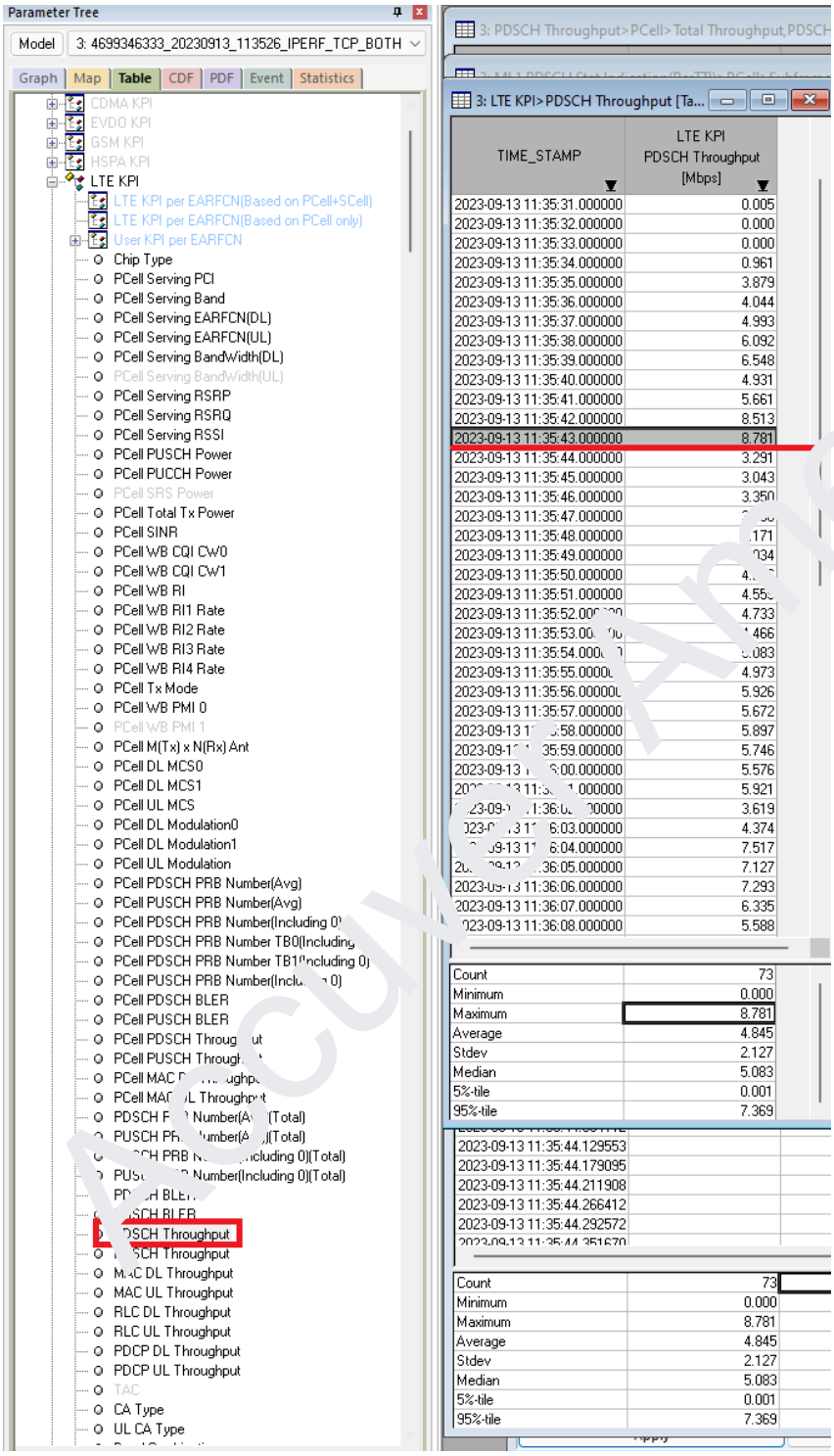
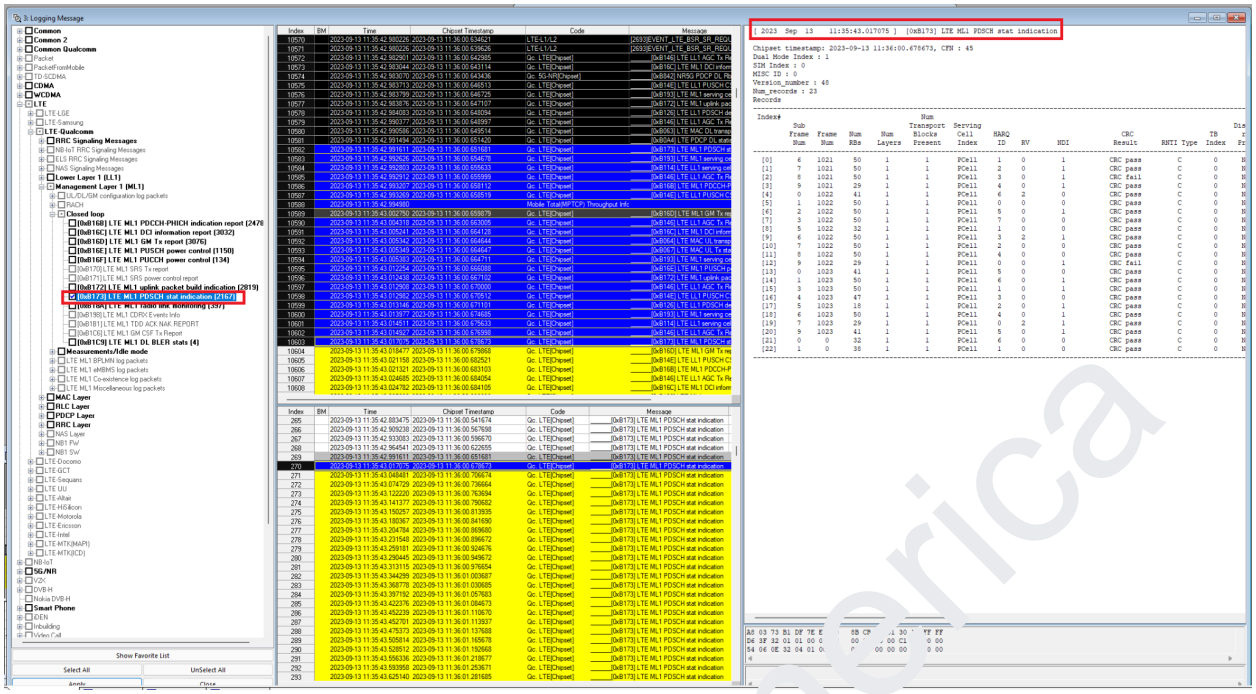
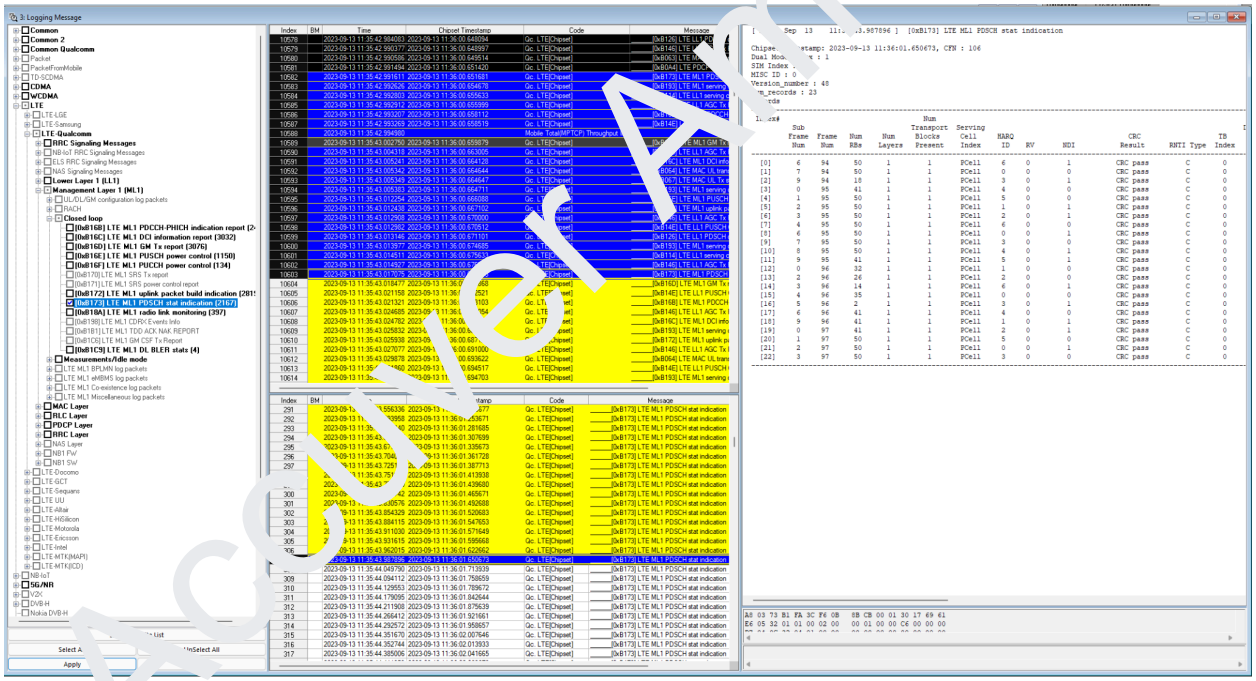
LTE PDSCH Throughput Calculation
Reference message: [0xB173] LTE ML1 PDSCH stat indication
Formula: PDSCH Throughput = Sum of TB Bytes / time duration
Example: Calculation at 11:35:43.000 PDSCH Throughput = 8.781Mbps



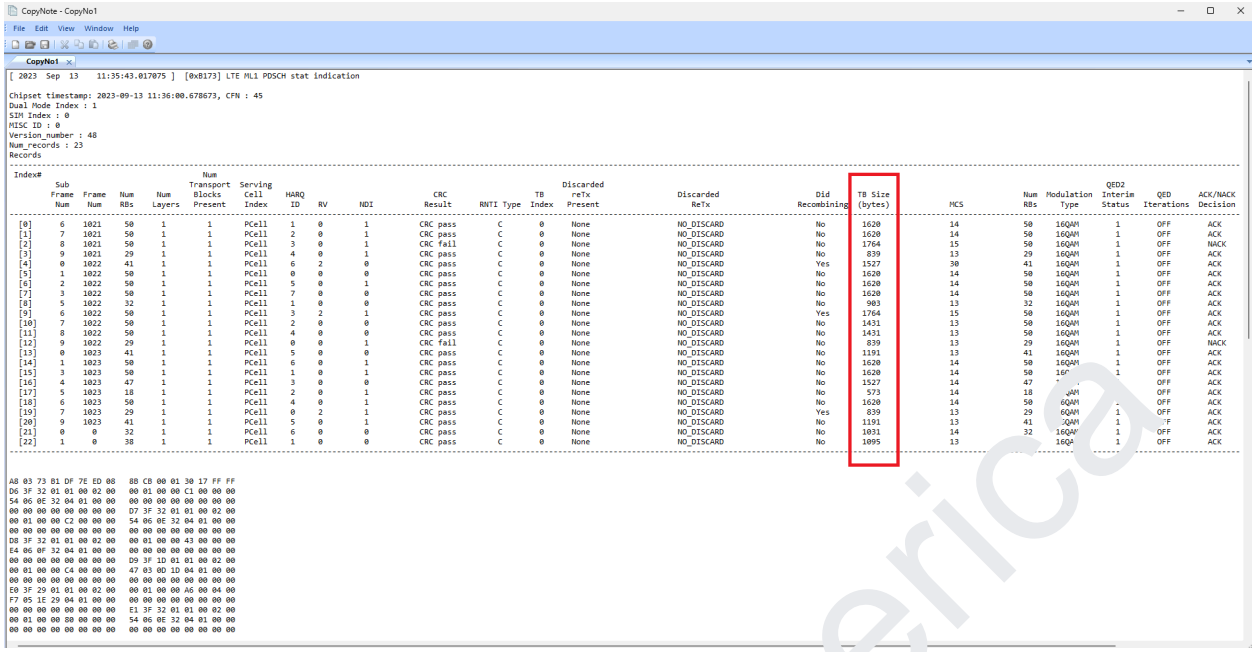
Reference messages from 11:35:43:017075(first message in 11:35:43) to 11:35:43:987896(last message in 11:35:43)





We are going to check the TB Size column

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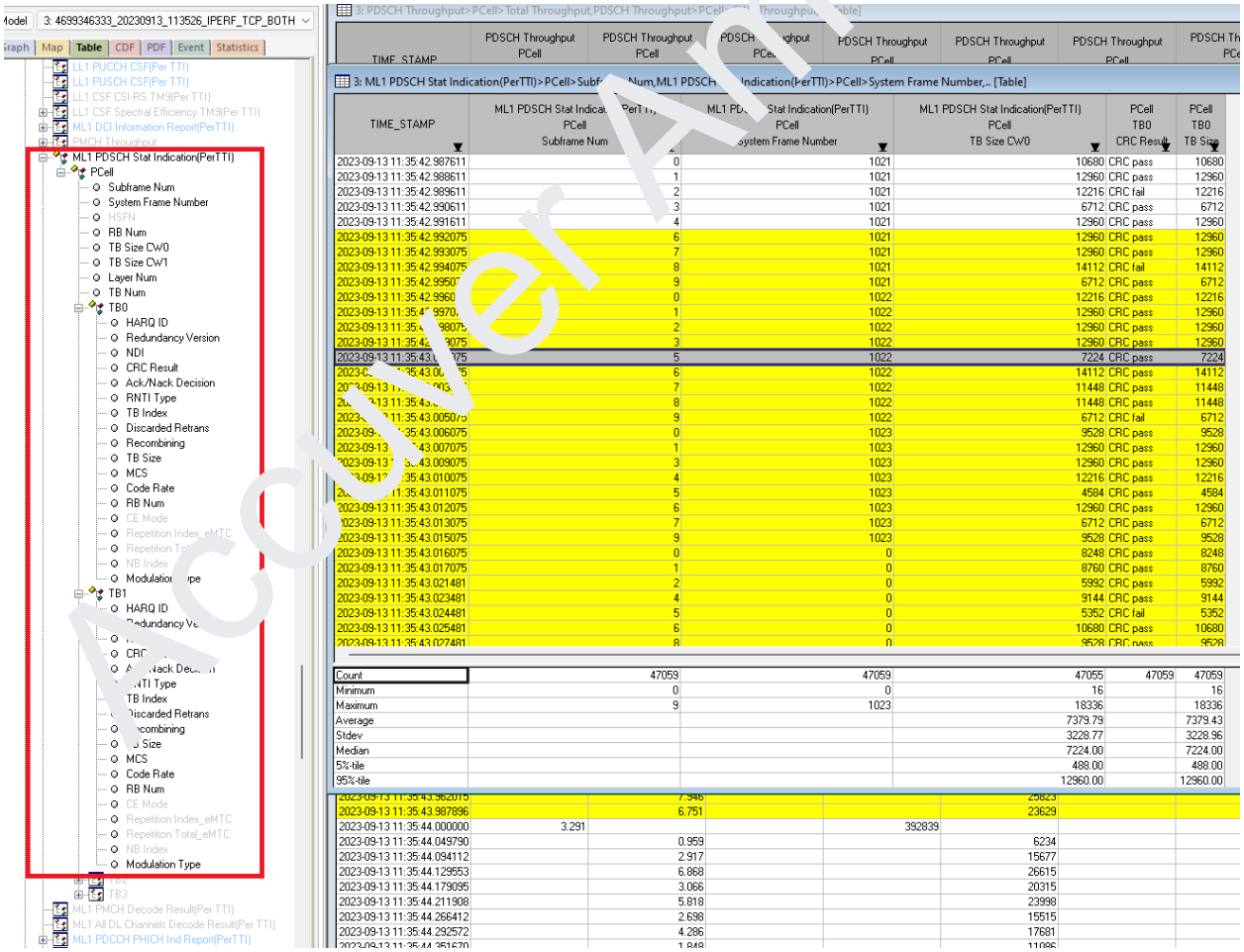
[2023 Sep 13 11:35:43.017075] [0x8173] LTE ML1 PDSCH stat indication

Chipset timestamp: 2023-09-13 11:36:00.678673, CFN : 45
Dual Mode Index : 1
SIM Index : 0
MISC ID : 0
Version_number : 48
Num_records : 23
Records

Index#	Sub Frame Num	Frame Num	Num RBs	Num Layers	Num Transport Blocks Present	Serving Cell Index	HARQ ID	RV	NDI	CRC Result	RNTI Type	TB Index	Discarded reTx Present	Discarded ReTx	Did Recombining	TB Size (bytes)	MCS	Num RBs	Modulation Type	QED2 Interim Status	QED Iterations	ACK/NACK Decision
[0]	6	1021	50	1	1	PCell	1	0	1	CRC pass	C	0	None	NO_DISCARD	No	1620	14	50	16QAM	1	OFF	ACK
[1]	7	1021	50	1	1	PCell	2	0	1	CRC pass	C	0	None	NO_DISCARD	No	1620	14	50	16QAM	1	OFF	ACK
[2]	8	1021	50	1	1	PCell	3	0	1	CRC fail	C	0	None	NO_DISCARD	No	1764	15	50	16QAM	1	OFF	NACK
[3]	9	1021	29	1	1	PCell	4	0	1	CRC pass	C	0	None	NO_DISCARD	No	839	13	29	16QAM	1	OFF	ACK
[4]	0	1022	41	1	1	PCell	6	2	0	CRC pass	C	0	None	NO_DISCARD	Yes	1527	30	41	16QAM	1	OFF	ACK
[5]	1	1022	50	1	1	PCell	0	0	0	CRC pass	C	0	None	NO_DISCARD	No	1620	14	50	16QAM	1	OFF	ACK
[6]	2	1022	50	1	1	PCell	5	0	1	CRC pass	C	0	None	NO_DISCARD	No	1620	14	50	16QAM	1	OFF	ACK
[7]	3	1022	50	1	1	PCell	7	0	0	CRC pass	C	0	None	NO_DISCARD	No	1620	14	50	16QAM	1	OFF	ACK
[8]	5	1022	32	1	1	PCell	1	0	0	CRC pass	C	0	None	NO_DISCARD	No	903	13	32	16QAM	1	OFF	ACK
[9]	6	1022	50	1	1	PCell	3	2	1	CRC pass	C	0	None	NO_DISCARD	Yes	1764	15	50	16QAM	1	OFF	ACK
[10]	7	1022	50	1	1	PCell	2	0	0	CRC pass	C	0	None	NO_DISCARD	No	1431	13	50	16QAM	1	OFF	ACK
[11]	8	1022	50	1	1	PCell	4	0	0	CRC pass	C	0	None	NO_DISCARD	No	1431	13	50	16QAM	1	OFF	ACK
[12]	9	1022	29	1	1	PCell	0	0	1	CRC fail	C	0	None	NO_DISCARD	No	839	13	29	16QAM	1	OFF	NACK
[13]	0	1023	41	1	1	PCell	5	0	0	CRC pass	C	0	None	NO_DISCARD	No	1191	13	41	16QAM	1	OFF	ACK
[14]	1	1023	50	1	1	PCell	6	0	1	CRC pass	C	0	None	NO_DISCARD	No	1620	14	50	16QAM	1	OFF	ACK
[15]	3	1023	50	1	1	PCell	1	0	1	CRC pass	C	0	None	NO_DISCARD	No	1620	14	50	16[illegible]	1	OFF	ACK
[16]	4	1023	47	1	1	PCell	3	0	0	CRC pass	C	0	None	NO_DISCARD	No	1527	14	47	[illegible]		OFF	ACK
[17]	5	1023	18	1	1	PCell	2	0	1	CRC pass	C	0	None	NO_DISCARD	No	573	14	18	[illegible]AM		OFF	ACK
[18]	6	1023	50	1	1	PCell	4	0	1	CRC pass	C	0	None	NO_DISCARD	No	1620	14	50	6QAM		OFF	ACK
[19]	7	1023	29	1	1	PCell	0	2	1	CRC pass	C	0	None	NO_DISCARD	Yes	839	13	29	6QAM	1	OFF	ACK
[20]	9	1023	41	1	1	PCell	5	0	1	CRC pass	C	0	None	NO_DISCARD	No	1191	13	41	6QAM	1	OFF	ACK
[21]	0	0	32	1	1	PCell	6	0	0	CRC pass	C	0	None	NO_DISCARD	No	1031	14	32	6QAM	1	OFF	ACK
[22]	1	0	38	1	1	PCell	1	0	0	CRC pass	C	0	None	NO_DISCARD	No	1095	13		6QA	1	OFF	ACK

A8 03 73 81 DF 7E ED 08 88 CB 00 01 30 17 FF FF
D6 3F 32 01 01 00 02 00 00 01 00 00 C1 00 00 00
54 06 0E 32 04 01 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 D7 3F 32 01 01 00 02 00
00 01 00 00 C2 00 00 00 54 06 0E 32 04 01 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D8 3F 32 01 01 00 02 00 00 01 00 00 43 00 00 00
E4 06 0F 32 04 01 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 D9 3F 1D 01 01 00 02 00
00 01 00 00 C4 00 00 00 47 03 0D 1D 04 01 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
E0 3F 29 01 01 00 02 00 00 01 00 00 A6 00 04 00
F7 05 1E 29 04 01 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 E1 3F 32 01 01 00 02 00
00 01 00 00 80 00 00 00 54 06 0E 32 04 01 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

ML1 PDSCH Stat Indication(PerTTI) will show every subfra[illegible] as one record.



3: 4693346333_20230913_113526_IPERF_TCP_BOTH

Graph | Map | Table | CDF | PDF | Event | Statistics

- LL1 PUCCH CSF(Per TTI)
- LL1 PUSCH CSF(Per TTI)
- LL1 CSF CSI-RS TM9(Per TTI)
- LL1 CSF Spectral Efficiency TM9(Per TTI)
- ML1 DCI Information Report(PerTTI)
- PMCH Throughput
- ML1 PDSCH Stat Indication(PerTTI)
  - PCell
    - Subframe Num
    - System Frame Number
    - HSFN
    - RB Num
    - TB Size CW0
    - TB Size CW1
    - Layer Num
    - TB Num
    - TB0
      - HARQ ID
      - Redundancy Version
      - NDI
      - CRC Result
      - Ack/Nack Decision
      - RNTI Type
      - TB Index
      - Discarded Retrans
      - Recombining
      - TB Size
      - MCS
      - Code Rate
      - RB Num
      - CE Mode
      - Repetition Index_eMTC
      - Repetition Total_eMTC
      - NB Index
      - Modulation Type
    - TB1
      - HARQ ID
      - [illegible]
      - Modulation Type
    - TB2
    - TB3
- ML1 PMCH Decode Result(Per TTI)
- ML1 All DL Channels Decode Result(Per TTI)
- ML1 PDCCH PHICH Ind Report(PerTTI)

3: PDSCH Throughput>PCell>Total Throughput,PDSCH Throughput>PCell>[illegible]Throughput,[illegible]Table]

TIME_STAMP	PDSCH Throughput PCell	PDSCH Throughput PCell	PDSCH [illegible]ghput PCell	PDSCH Throughput PCell	PDSCH Throughput PCell	PDSCH Th PCe

3: ML1 PDSCH Stat Indication(PerTTI)>PCell>Subf[illegible] Num,ML1 PDSCH [illegible]Indication(PerTTI)>PCell>System Frame Number,.. [Table]

TIME_STAMP	ML1 PDSCH Stat Indica[illegible] PCell Subframe Num	[illegible]PerTTI)	ML1 PD[illegible] Stat Indication(PerTTI) PCell System Frame Number	ML1 PDSCH Stat Indication(PerTTI) PCell TB Size CW0	PCell TB0 CRC Result	PCell TB0 TB Size
2023-09-13 11:35:42.987611	0		1021	10680	CRC pass	10680
2023-09-13 11:35:42.988611	1		1021	12960	CRC pass	12960
2023-09-13 11:35:42.989611	2		1021	12216	CRC fail	12216
2023-09-13 11:35:42.990611	3		1021	6712	CRC pass	6712
2023-09-13 11:35:42.991611	4		1021	12960	CRC pass	12960
2023-09-13 11:35:42.992075	6		1021	12960	CRC pass	12960
2023-09-13 11:35:42.993075	7		1021	12960	CRC pass	12960
2023-09-13 11:35:42.994075	8		1021	14112	CRC fail	14112
2023-09-13 11:35:42.99507[illegible]	9		1021	6712	CRC pass	6712
2023-09-13 11:35:42.9960[illegible]	0		1022	12216	CRC pass	12216
2023-09-13 11:35:4[illegible].9970[illegible]	1		1022	12960	CRC pass	12960
2023-09-13 11:35:4[illegible].9980[illegible]	2		1022	12960	CRC pass	12960
2023-09-13 11:35:42[illegible].0075	3		1022	12960	CRC pass	12960
2023-09-13 11:35:43.0[illegible]075	5		1022	7224	CRC pass	7224
2023-0[illegible]:35:43.00[illegible]75	6		1022	14112	CRC pass	14112
2023-09-13 11:35:43.[illegible]003	7		1022	11448	CRC pass	11448
2023-09-13 11:35:43.[illegible]	8		1022	11448	CRC pass	11448
2023-[illegible]:35:43.0050[illegible]	9		1022	6712	CRC fail	6712
2023-09-[illegible]:35:43.006075	0		1023	9528	CRC pass	9528
2023-09-13 [illegible]:35:43.007075	1		1023	12960	CRC pass	12960
2023-09-13 [illegible]:35:43.009075	3		1023	12960	CRC pass	12960
2023-09-[illegible]:35:43.010075	4		1023	12216	CRC pass	12216
2023-[illegible]:35:43.011075	5		1023	4584	CRC pass	4584
2023-09-13 11:35:43.012075	6		1023	12960	CRC pass	12960
2023-09-13 11:35:43.013075	7		1023	6712	CRC pass	6712
2023-09-13 11:35:43.015075	9		1023	9528	CRC pass	9528
2023-09-13 11:35:43.016075	0		0	8248	CRC pass	8248
2023-09-13 11:35:43.017075	1		0	8760	CRC pass	8760
2023-09-13 11:35:43.021481	2		0	5992	CRC pass	5992
2023-09-13 11:35:43.023481	4		0	9144	CRC pass	9144
2023-09-13 11:35:43.024481	5		0	5352	CRC fail	5352
2023-09-13 11:35:43.025481	6		0	10680	CRC pass	10680
2023-09-13 11:35:43.027481	8		0	9528	CRC pass	9528

Count	47059	47059	47055	47059	47059	
Minimum	0	0	16		16	
Maximum	9	1023	18336		18336	
Average			7379.79		7379.43	
Stdev			3228.77		3228.96	
Median			7224.00		7224.00	
5%-tile			488.00		488.00	
95%-tile			12960.00		12960.00	

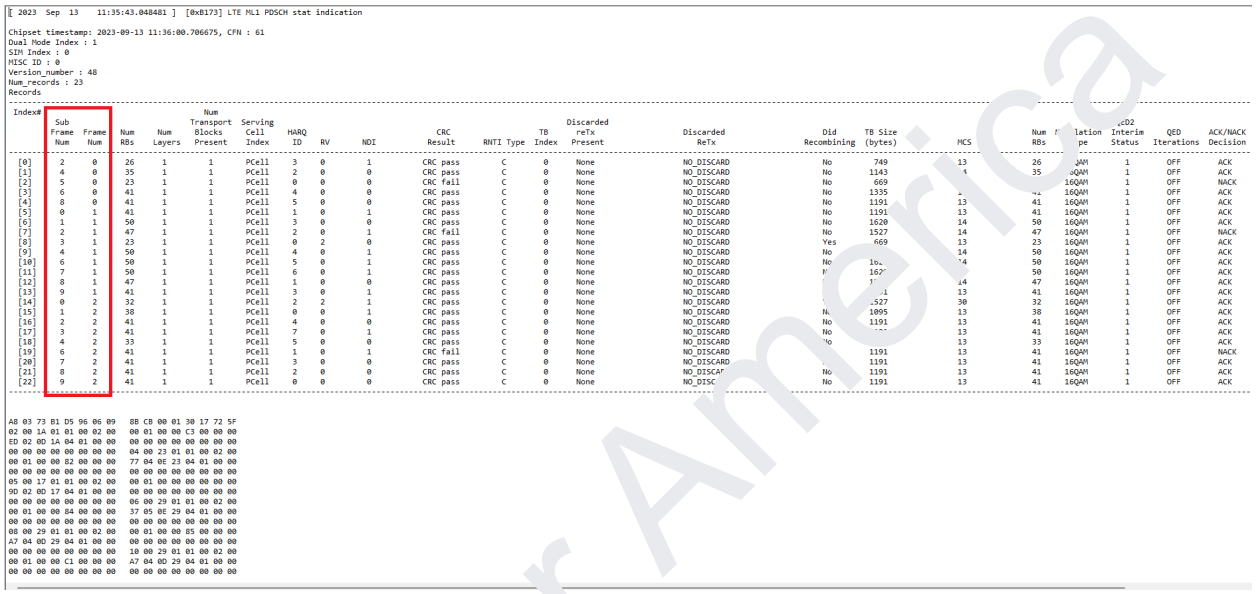
2023-09-13 11:35:43.362075		7.946		25823		
2023-09-13 11:35:43.987896		6.751		23629		
2023-09-13 11:35:44.000000	3.291		392839			
2023-09-13 11:35:44.049790		0.959		6234		
2023-09-13 11:35:44.094112		2.917		15677		
2023-09-13 11:35:44.129553		6.868		26615		
2023-09-13 11:35:44.179095		3.066		20315		
2023-09-13 11:35:44.211908		5.818		23998		
2023-09-13 11:35:44.266412		2.698		15515		
2023-09-13 11:35:44.292572		4.286		17681		
2023-09-13 11:35:44.351670		1.848		11096		

Sum all the values in TB Size(we don't have CW1 in this second, so only CW0 values) =

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12960+12960+14112+.... = 8631432 bits
Note: the TB Size value is in bits so it equals the TB Size(bytes) * 8 bits/bytes in 0xb173 raw message

Now we need to calculate the time duration. LTE configures system frame = 10ms, subframe = 1ms
First we need to calculate the time duration of each 0xb173 message. For example the below one, the duration =
Frame2 subframe9 - Frame0 subframe2 = (2*10ms+9*1ms) - (0*10ms+2*1ms) = 27ms



[2023 Sep 13 11:35:43.048481] [0xB173] LTE ML1 PDSCH stat indication

Chipset timestamp: 2023-09-13 11:36:00.706675, CFN : 61
Dual Mode Index : 1
SIM Index : 0
MISC ID : 0
Version_number : 48
Num_records : 23
Records

Index#	Sub Frame Num	Frame Num	Num RBs	Num Layers	Num Transport Blocks Present	Serving Cell Index	HARQ ID	RV	NDI	CRC Result	RNTI Type	TB Index	Discarded reTx Present	Discarded ReTx	Did Recombining	TB Size (bytes)	MCS	Num RBs	Modulation Type	2cD2 Interim Status	QED Iterations	ACK/NACK Decision
[0]	2	0	26	1	1	PCell	3	0	1	CRC pass	C	0	None	NO_DISCARD	No	749	13	26	[illegible]AM	1	OFF	ACK
[1]	4	0	35	1	1	PCell	2	0	0	CRC pass	C	0	None	NO_DISCARD	No	1143	[illegible]	35	[illegible]QAM	1	OFF	ACK
[2]	5	0	23	1	1	PCell	0	0	0	CRC fail	C	0	None	NO_DISCARD	No	669	[illegible]	[illegible]	16QAM	1	OFF	NACK
[3]	6	0	41	1	1	PCell	4	0	0	CRC pass	C	0	None	NO_DISCARD	No	1335	[illegible]	[illegible]	16QAM	1	OFF	ACK
[4]	8	0	41	1	1	PCell	5	0	0	CRC pass	C	0	None	NO_DISCARD	No	1191	13	41	16QAM	1	OFF	ACK
[5]	0	1	41	1	1	PCell	1	0	1	CRC pass	C	0	None	NO_DISCARD	No	1191	13	41	16QAM	1	OFF	ACK
[6]	1	1	50	1	1	PCell	3	0	0	CRC pass	C	0	None	NO_DISCARD	No	1620	14	50	16QAM	1	OFF	ACK
[7]	2	1	47	1	1	PCell	2	0	1	CRC fail	C	0	None	NO_DISCARD	No	1527	14	47	16QAM	1	OFF	NACK
[8]	3	1	23	1	1	PCell	0	2	0	CRC pass	C	0	None	NO_DISCARD	Yes	669	13	23	16QAM	1	OFF	ACK
[9]	4	1	50	1	1	PCell	4	0	1	CRC pass	C	0	None	NO_DISCARD	No	[illegible]	14	50	16QAM	1	OFF	ACK
[10]	6	1	50	1	1	PCell	5	0	1	CRC pass	C	0	None	NO_DISCARD	No	[illegible]	[illegible]4	50	16QAM	1	OFF	ACK
[11]	7	1	50	1	1	PCell	6	0	1	CRC pass	C	0	None	NO_DISCARD	N[illegible]	162[illegible]	[illegible]	50	16QAM	1	OFF	ACK
[12]	8	1	47	1	1	PCell	1	0	0	CRC pass	C	0	None	NO_DISCARD	[illegible]	[illegible]	[illegible]4	47	16QAM	1	OFF	ACK
[13]	9	1	41	1	1	PCell	3	0	1	CRC pass	C	0	None	NO_DISCARD	[illegible]	[illegible]1	13	41	16QAM	1	OFF	ACK
[14]	0	2	32	1	1	PCell	2	2	1	CRC pass	C	0	None	NO_DISCARD	[illegible]	[illegible]527	30	32	16QAM	1	OFF	ACK
[15]	1	2	38	1	1	PCell	0	0	1	CRC pass	C	0	None	NO_DISCARD	No	1095	13	38	16QAM	1	OFF	ACK
[16]	2	2	41	1	1	PCell	4	0	0	CRC pass	C	0	None	NO_DISCARD	No	1191	13	41	16QAM	1	OFF	ACK
[17]	3	2	41	1	1	PCell	7	0	1	CRC pass	C	0	None	NO_DISCARD	No	[illegible]	13	41	16QAM	1	OFF	ACK
[18]	4	2	33	1	1	PCell	5	0	0	CRC pass	C	0	None	NO_DISCARD	[illegible]o	[illegible]	13	33	16QAM	1	OFF	ACK
[19]	6	2	41	1	1	PCell	1	0	1	CRC fail	C	0	None	NO_DISCARD	[illegible]	1191	13	41	16QAM	1	OFF	NACK
[20]	7	2	41	1	1	PCell	3	0	0	CRC pass	C	0	None	NO_DISCARD	[illegible]	1191	13	41	16QAM	1	OFF	ACK
[21]	8	2	41	1	1	PCell	2	0	0	CRC pass	C	0	None	NO_DISCA[illegible]	No	1191	13	41	16QAM	1	OFF	ACK
[22]	9	2	41	1	1	PCell	0	0	0	CRC pass	C	0	None	NO_DISC[illegible]	No	1191	13	41	16QAM	1	OFF	ACK

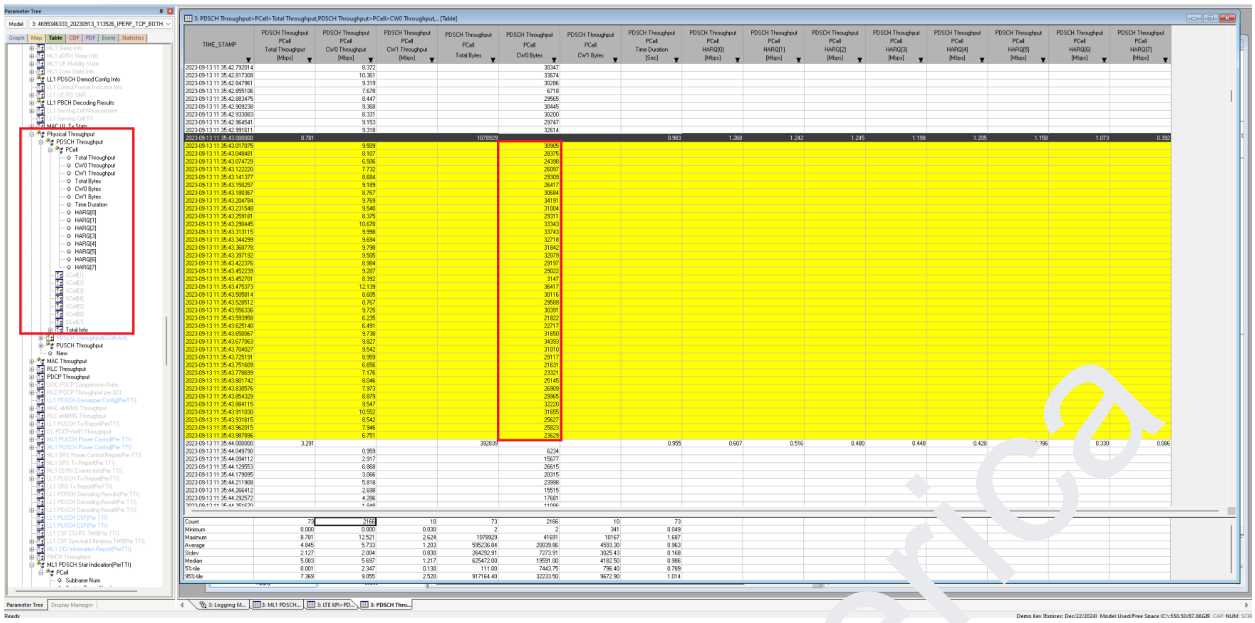
A8 03 73 B1 D5 96 06 09 8B CB 00 01 30 17 72 5F
02 00 1A 01 01 00 02 00 00 01 00 00 C3 00 00 00
ED 02 0D 1A 04 01 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 04 00 23 01 01 00 02 00
00 01 00 00 82 00 00 00 77 04 0E 23 04 01 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
05 00 17 01 01 00 02 00 00 01 00 00 00 00 00 00
9D 02 0D 17 04 01 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 06 00 29 01 01 00 02 00
00 01 00 00 84 00 00 00 37 05 0E 29 04 01 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
08 00 29 01 01 00 02 00 00 01 00 00 85 00 00 00
A7 04 0D 29 04 01 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 10 00 29 01 01 00 02 00
00 01 00 00 C1 00 00 00 A7 04 0D 29 04 01 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

We sum up all the duration values for [illegible] 0x[illegible]173 message within this second.
It's 983 ms.

Finally, we can get the result.
PDSCH Throughput
= Sum of TB Bytes / time dura[illegible]n
= 8631432 bits / 98[illegible]ms
= 8780.703967 bits/ms
= 8.781 Mbps

If you want to check summary information for each 0xb173 message, the KPI below is also helpful.
LTE/LTE-Adv Qualcomm -> Physical Throughput->PCell(Total) Info

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